Reading Graphs

Name _____ Date _____ No calculator Version 1

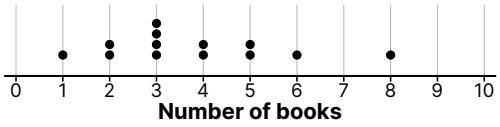
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KEY WORDS dot plot *(kauwhata ira)* · bar graph *(kauwhata pae)* · time-series graph *(kauwhata raupapa wā)*

A Dot plots ●○○

Remember: Each dot is one piece of data. The mode is the value with the most dots.

1. The dot plot shows the number of books read by each student in a group last month.



a) How many students are in the group? _____

b) What is the mode? _____

c) What is the largest number of books read? _____

d) How many students read more than 4 books? _____

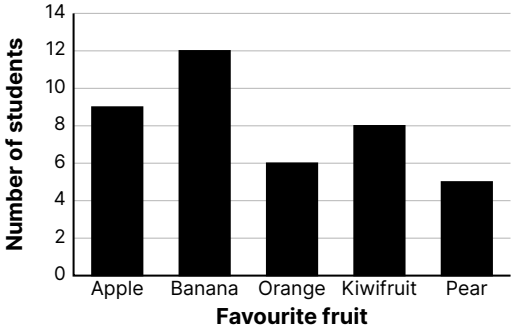
e) What is the range of the number of books? _____

f) How many books did the group read altogether? _____

B Bar graphs ●○○

Remember: The height of each bar shows how many are in that group. Read the height from the scale on the side.

2. The bar graph shows the favourite fruit of the students in a class.



a) Which fruit is the most popular? _____

b) How many students chose orange? _____

c) How many students are in the class? _____

d) How many more chose banana than pear? _____

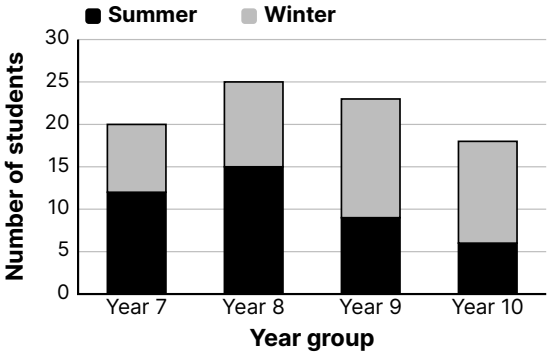
e) How many students did not choose apple? _____

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C Stacked bar graphs ●●○

Remember: In a stacked bar graph each bar is split into parts. The key shows what each part stands for. The whole height of a bar is the total.

3. The stacked bar graph shows how many students from each year group play a summer sport and a winter sport.



a) How many Year 8 students play a summer sport?

b) How many Year 8 students play a winter sport?

c) How many Year 8 students are shown altogether?

d) Which year group has the most students altogether?

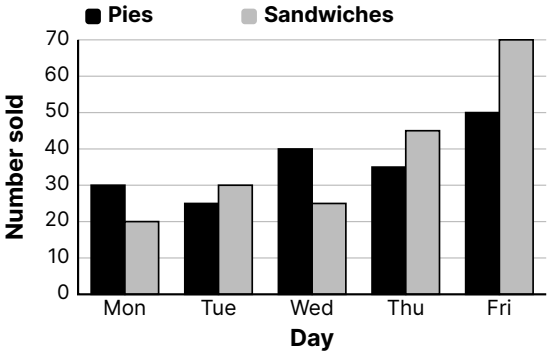
e) How many students play a winter sport altogether?

f) In which year groups do more students play a winter sport than a summer sport?

D Clustered bar graphs ●●○

Remember: A clustered bar graph puts the bars for two groups side by side, so they are easy to compare. Use the key to see which bar is which.

4. The clustered bar graph shows the number of pies and sandwiches sold at a school canteen each day for a week.



a) How many pies were sold on Wed?

b) How many sandwiches were sold on Wed?

c) On which days were more sandwiches sold than pies?

d) On which day was the difference between pies and sandwiches the greatest?

e) How many pies were sold in the whole week?

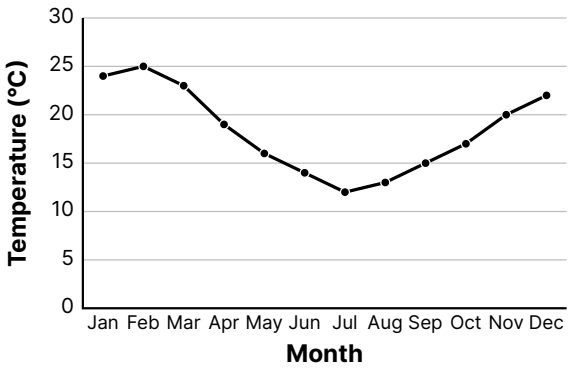
f) Which sold more over the whole week, pies or sandwiches? By how many?

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E Time-series graphs ●●●

Remember: A time-series graph shows how something changes over time. Look for the overall trend as well as the ups and downs.

5. The graph shows the average daily high temperature in a New Zealand town for each month of one year.



a) What was the temperature in Apr?

b) Which month was the warmest?

c) Which month was the coldest?

d) What is the difference between the warmest and the coldest months?

e) Between which two months next to each other did the temperature fall the most?

f) In how many months was the temperature above 20°C?

g) Describe how the temperature changes over the year.

Extension

6. Look again at the stacked bar graph. What fraction of all the students shown play a summer sport? Give your answer in its simplest form.

7. Give one reason a time-series graph is a better choice than a bar graph for showing the temperature over a year.

8. The same data could be shown as a clustered bar graph. Give one thing that is easier to see on a stacked bar graph, and one thing that is easier to see on a clustered bar graph.

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Reading Graphs

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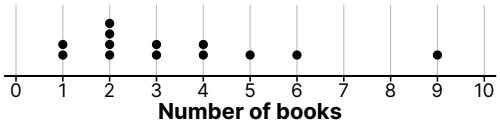
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KEY WORDS dot plot *(kauwhata ira)* · bar graph *(kauwhata pae)* · time-series graph *(kauwhata raupapa wā)*

A Dot plots ●○○

Remember: Each dot is one piece of data. The mode is the value with the most dots.

1. The dot plot shows the number of books read by each student in a group last month.



a) How many students are in the group? _____

b) What is the mode? _____

c) What is the largest number of books read? _____

d) How many students read more than 3 books? _____

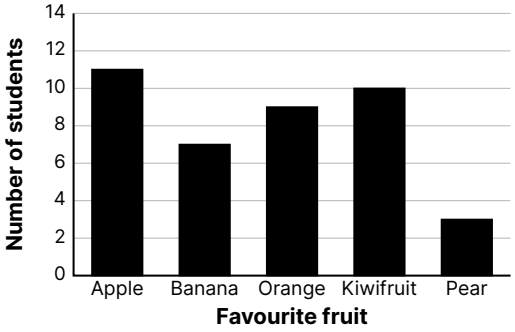
e) What is the range of the number of books? _____

f) How many books did the group read altogether? _____

B Bar graphs ●○○

Remember: The height of each bar shows how many are in that group. Read the height from the scale on the side.

2. The bar graph shows the favourite fruit of the students in a class.



a) Which fruit is the most popular? _____

b) How many students chose orange? _____

c) How many students are in the class? _____

d) How many more chose apple than pear? _____

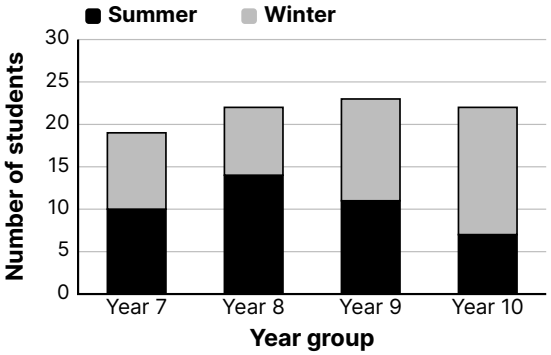
e) How many students did not choose apple? _____

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C Stacked bar graphs ●●○

Remember: In a stacked bar graph each bar is split into parts. The key shows what each part stands for. The whole height of a bar is the total.

3. The stacked bar graph shows how many students from each year group play a summer sport and a winter sport.



a) How many Year 9 students play a summer sport?

b) How many Year 9 students play a winter sport?

c) How many Year 9 students are shown altogether?

d) Which year group has the most students altogether?

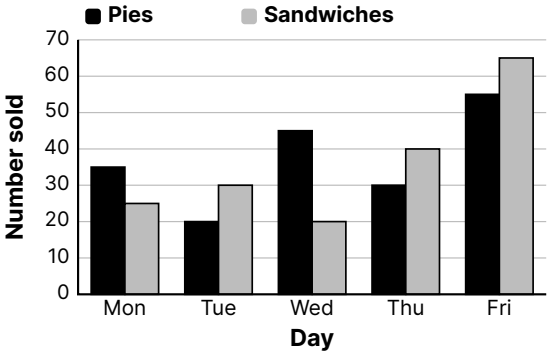
e) How many students play a winter sport altogether?

f) In which year groups do more students play a winter sport than a summer sport?

D Clustered bar graphs ●●○

Remember: A clustered bar graph puts the bars for two groups side by side, so they are easy to compare. Use the key to see which bar is which.

4. The clustered bar graph shows the number of pies and sandwiches sold at a school canteen each day for a week.



a) How many pies were sold on Thu?

b) How many sandwiches were sold on Thu?

c) On which days were more sandwiches sold than pies?

d) On which day was the difference between pies and sandwiches the greatest?

e) How many pies were sold in the whole week?

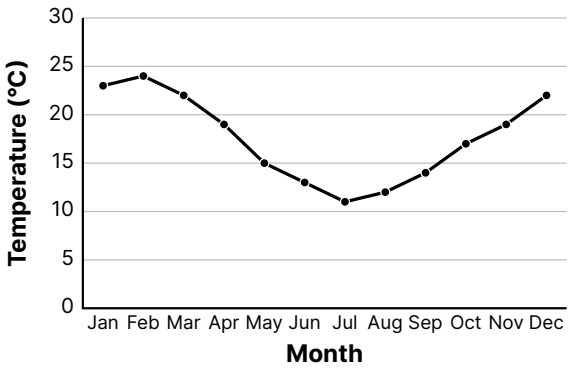
f) Which sold more over the whole week, pies or sandwiches? By how many?

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E Time-series graphs ●●●

Remember: A time-series graph shows how something changes over time. Look for the overall trend as well as the ups and downs.

5. The graph shows the average daily high temperature in a New Zealand town for each month of one year.



a) What was the temperature in May?

b) Which month was the warmest?

c) Which month was the coldest?

d) What is the difference between the warmest and the coldest months?

e) Between which two months next to each other did the temperature fall the most?

f) In how many months was the temperature above 19°C?

g) Describe how the temperature changes over the year.

Extension

6. Look again at the stacked bar graph. What fraction of all the students shown play a summer sport? Give your answer in its simplest form.

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